Pier Review

Parallel parking for warships, powered by physics

THE OPERATIONAL CHALLENGE

Drafting a safe mooring and pier-work plan is complex. Bridge teams often rely on static sketches and rough estimates that cannot readily reflect changing wind, current, tide or mooring geometry.

STATEMENT OF OBJECTIVES

Give bridge teams dynamic situational awareness to replace guesswork with calculated loads, compare configurations and prepare a shared, actionable plan for safer pier-work execution.

HOW WE SOLVE IT

Interactive planning + automated physics

Pier Review replaces static manual sketches with a dynamic workspace that calculates bollard loads, line tensions and environmental forces as the plan changes.

- **NOAA data + NSTM 582 checks**
  Use NOAA data where available to assess geometry and loads against NSTM 582 (Mooring and Towing) referenced limits.
- **Structured mooring sequences**
  Draft line-handling sequences that support coordinated operational readiness.
- **Automatic safety sheets**
  Generate comprehensive Mooring Evaluation Safety Sheets for review and briefing.
- **Efficient tug placement**
  Configure tug positions and assistance to support precise maneuvering.
- **Intuitive graphical editing**
  Drag and drop the ship, then edit mooring connections to compare plan configurations.
- **Mapbox visual geography**
  Explore the berth and surrounding area in a rich, interactive geographic view.

APPLICATION SCREENSHOT

Mooring layout, forces & tug placement

APPLICATION SCREENSHOT

Line sequence & Mooring Evaluation Safety Sheet

OPERATIONAL USE CASE

Before arrival or pier work, select a berth, place the ship and review wind, current and tide. Compare tug and line configurations, then brief the selected sequence and safety sheet. Reassess the plan when conditions change.

TARGET AUDIENCE

Primary: bridge teams, deck officers, harbor pilots and mooring crews.

Supporting: tug coordinators, port operations and fleet readiness planners.

FIELDING REQUIREMENTS

Proposed deployment configuration

Aligned with platforms already being deployed to ships, submarines and autonomous vehicles.

Kubernetes platform
Domino, UDS or vanilla Kubernetes to host and manage the application services.

Dedicated GPU hardware
Hardware acceleration for real-time physics calculation pods.

Single sign-on
Keycloak or enterprise SSO for identity authentication and access control.

Engineering decision support. The current demonstrator uses seed data; operational fielding requires qualified inputs and human review.